GDP and Emissions Database

Version 1.0

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Data is documented in the following table structure:

Variable	Range	Description
Name in Dataset	FirstObs–LastObs	Title; Unit Construction, Source

The datasets are provided both as individual `.csv` files and as a consolidated `.xlsx` file. The `.csv` files are titled `cc_data.csv`, where `cc` is the two-letter country code in lowercase. Each `.csv` file uses a comma as the value separator, a dot as the decimal separator, and the first row as the header. The consolidated `.xlsx` file contains the data organized in Excel sheets titled `CC`, where `CC` is the two-letter country code in uppercase.

Contents

1. Australia (AU)

Variable	Range	Description
gdp	1950–2023	Real GDP; 2017 AUD, millions The following series rescaled to match the value of the series Australian Bureau of Statistics (2026*e*) in 2017: i) Since 1959: Australian Bureau of Statistics (2026*d*) ii) Prior to 1959: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2023	Real net capital stock; 2017 AUD, millions The following series rescaled to match the value of the series Australian Bureau of Statistics (2026*c*) in 2017: i) Since 1959: Australian Bureau of Statistics (2026*b*) ii) Prior to 1959: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1994: OECD (2026*b*) ii) Prior to 1994: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions i) Since 1994: OECD (2026*c*) ii) Prior to 1994: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1860–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1959: Share of Australian Bureau of Statistics (2026*a*) in the total of Australian Bureau of Statistics (2026*a*) and Australian Bureau of Statistics (2026*f*) ii) Prior to 1959: Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

2. Austria (AT)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026*d*) in 2017: i) Since 1995: Eurostat (2026*c*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026*h*) in 2017: i) Since 1995: Eurostat (2026*g*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026*b*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026*f*) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1819–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026*a*) in the total of Eurostat (2026*a*) and Eurostat (2026*e*) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

3. Belgium (BE)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026*d*) in 2017: i) Since 1995: Eurostat (2026*c*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026*h*) in 2017: i) Since 1995: Eurostat (2026*g*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026*b*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026*f*) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1830–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026*a*) in the total of Eurostat (2026*a*) and Eurostat (2026*e*) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

4. Canada (CA)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 CAD, millions The following series rescaled to match the value of the series Statistics Canada (2026*e*) in 2017: i) Since 1981: Statistics Canada (2026*d*) ii) Prior to 1981: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 CAD, millions The following series rescaled to match the end of year value of the series Statistics Canada (2026*c*) in 2017: i) Since 2009: End of year values of Statistics Canada (2026*b*) ii) Prior to 2009: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1970: OECD (2026*b*) ii) Prior to 1970: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions i) Since 1997: OECD (2026*c*) ii) Prior to 1997: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1785–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1981: Share of Statistics Canada (2026*a*) in the total of Statistics Canada (2026*a*) and Statistics Canada (2026*f*) ii) Prior to 1981: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

5. Denmark (DK)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 DKK, millions The following series rescaled to match the value of the series Eurostat (2026*d*) in 2017: i) Since 1975: Eurostat (2026*c*) ii) Prior to 1975: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 DKK, millions The following series rescaled to match the end of year value of the series Eurostat (2026*h*) in 2017: i) Since 1975: Eurostat (2026*g*) ii) Prior to 1975: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1975: Eurostat (2026*b*) ii) Prior to 1975: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1975: Eurostat (2026*f*) ii) Prior to 1975: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1843–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026*a*) in the total of Eurostat (2026*a*) and Eurostat (2026*e*) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

6. Finland (FI)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026*d*) in 2017: i) Since 1980: Eurostat (2026*c*) ii) Prior to 1980: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026*h*) in 2017: i) Since 1980: Eurostat (2026*g*) ii) Prior to 1980: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1980: Eurostat (2026*b*) ii) Prior to 1980: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1980: Eurostat (2026*f*) ii) Prior to 1980: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1860–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1975: Share of Eurostat (2026*a*) in the total of Eurostat (2026*a*) and Eurostat (2026*e*) ii) Prior to 1975: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

7. France (FR)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026*d*) in 2017: i) Since 1975: Eurostat (2026*c*) ii) Prior to 1975: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026*h*) in 2017: i) Since 1978: Eurostat (2026*g*) ii) Prior to 1978: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1978: Eurostat (2026*b*) ii) Prior to 1978: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1978: Eurostat (2026*f*) ii) Prior to 1978: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1810–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1949–2024	Labor share Share of Eurostat (2026*a*) in the total of Eurostat (2026*a*) and Eurostat (2026*e*)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

8. Germany (DE)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026*d*) in 2017: i) Since 1991: Eurostat (2026*c*) ii) Prior to 1991: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)

Variable	Range	Description
nks	1950–2024	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026*h*) in 2017: i) Since 1995: Eurostat (2026*g*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026*b*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026*f*) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1945–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026*a*) in the total of Eurostat (2026*a*) and Eurostat (2026*e*)

Variable	Range	Description
		ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

9. Greece (GR)

Variable	Range	Description
gdp	1951–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026*d*) in 2017: i) Since 1995: Eurostat (2026*c*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1951–2023	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026*h*) in 2017: i) Since 1995: Eurostat (2026*g*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)

Variable	Range	Description
emp	1951–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026*b*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1951–2024	Hours worked; millions The following series divided by 1000:s i) Since 1995: Eurostat (2026*f*) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1913–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1951–2024	Labor share i) Since 1995: Share of Eurostat (2026*a*) in the total of Eurostat (2026*a*) and Eurostat (2026*e*) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1951–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017

Variable	Range	Description
tfph	1951–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1951–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

10. Ireland (IE)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026*d*) in 2017: i) Since 1995: Eurostat (2026*c*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2023	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026*h*) in 2017: i) Since 1995: Eurostat (2026*g*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026*b*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)

Variable	Range	Description
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026*f*) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1924–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026*a*) in the total of Eurostat (2026*a*) and Eurostat (2026*e*) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026)

Variable	Range	Description
		ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

11. Italy (IT)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026*d*) in 2017: i) Since 1995: Eurostat (2026*c*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026*h*) in 2017: i) Since 1995: Eurostat (2026*g*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026*b*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026*f*) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)

Variable	Range	Description
co2	1860–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026*a*) in the total of Eurostat (2026*a*) and Eurostat (2026*e*) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

12. Japan (JP)

Variable	Range	Description
gdp	1950–2023	Real GDP; 2017 JPY, millions The following series multiplied by 1000 after being rescaled to match the value of the series Statistics Bureau of Japan (2026*c*) in 2017: i) Since 1994: Statistics Bureau of Japan (2026*b*) ii) Prior to 1994: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2023	Real net capital stock; 2017 JPY, millions The following series multiplied by 1000 after being rescaled to match the end of year value of the series Statistics Bureau of Japan (2026*f*) in 2017: i) Since 1994: Statistics Bureau of Japan (2026*e*) ii) Prior to 1994: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1970: OECD (2026*b*) ii) Prior to 1970: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions i) Since 1970: OECD (2026*c*) ii) Prior to 1970: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1950–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2023	Labor share i) Since 1994: Share of Statistics Bureau of Japan (2026*a*) in the total of Statistics Bureau of Japan (2026*a*) and Statistics Bureau of Japan (2026*d*) ii) Prior to 1994: Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

13. Luxembourg (LU)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026*d*) in 2017: i) Since 1995: Eurostat (2026*c*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2023	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026*h*) in 2017: i) Since 1995: Eurostat (2026*g*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026*b*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1970–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026*f*) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1945–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026*a*) in the total of Eurostat (2026*a*) and Eurostat (2026*e*) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1970–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

14. Netherlands (NL)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026*d*) in 2017: i) Since 1995: Eurostat (2026*c*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026*h*) in 2017: i) Since 1995: Eurostat (2026*g*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026*b*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1970–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026*f*) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1846–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026*a*) in the total of Eurostat (2026*a*) and Eurostat (2026*e*) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1970–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

15. New Zealand (NZ)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 NZD, millions The following series rescaled to match the value of the series Statistics New Zealand (2026*c*) in 2017: i) Since 1987: Statistics New Zealand (2026*b*) ii) Prior to 1987: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 NZD, millions The following series rescaled to match the end of year value of the series Statistics New Zealand (2026*f*) in 2017: i) Since 1971: Statistics New Zealand (2026*e*) ii) Prior to 1971: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1989: OECD (2026*b*) ii) Prior to 1989: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1970–2024	Hours worked; millions i) Since 1971: OECD (2026*c*) ii) Prior to 1971: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1878–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)

Variable	Range	Description
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2023	Labor share i) Since 1971: Share of Statistics New Zealand (2026*a*) in the total of Statistics New Zealand (2026*a*) and Statistics New Zealand (2026*d*) ii) Prior to 1971: i) backdated using growth rates (previous period) of Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1970–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017

16. Norway (NO)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 NOK, millions The following series rescaled to match the value of the series Statistics Norway (2026*d*) in 2017: i) Since 1970: Statistics Norway (2026*c*) ii) Prior to 1970: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)

Variable	Range	Description
nks	1950–2024	Real net capital stock; 2017 NOK, millions The following series rescaled to match the end of year value of the series Statistics Norway (2026*f*) in 2017: i) Since 1970: Statistics Norway (2026*e*) ii) Prior to 1970: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1970: OECD (2026*b*) ii) Prior to 1970: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions i) Since 1970: OECD (2026*c*) ii) Prior to 1970: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1835–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2023	Labor share i) Since 1970: Share of Statistics Norway (2026*a*) in the total of Statistics Norway (2026*a*), Statistics Norway (2026*g*) and Statistics Norway (2026*b*)

Variable	Range	Description
		ii) Prior to 1970: Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

17. Portugal (PT)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026*d*) in 2017: i) Since 1995: Eurostat (2026*c*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2023	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026*h*) in 2017: i) Since 2000: Eurostat (2026*g*) ii) Prior to 2000: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)

Variable	Range	Description
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026*b*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026*f*) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1870–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026*a*) in the total of Eurostat (2026*a*) and Eurostat (2026*e*) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017

Variable	Range	Description
tfph	1950–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

18. Spain (ES)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 EUR, millions The following series rescaled to match the value of the series Eurostat (2026*d*) in 2017: i) Since 1995: Eurostat (2026*c*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2023	Real net capital stock; 2017 EUR, millions The following series rescaled to match the end of year value of the series Eurostat (2026*h*) in 2017: i) Since 1995: Eurostat (2026*g*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1995: Eurostat (2026*b*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)

Variable	Range	Description
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1995: Eurostat (2026*f*) ii) Prior to 1995: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1830–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026*a*) in the total of Eurostat (2026*a*) and Eurostat (2026*e*) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2023	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2023	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026)

Variable	Range	Description
		ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

19. Sweden (SE)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 SEK, millions The following series rescaled to match the value of the series Eurostat (2026*d*) in 2017: i) Since 1993: Eurostat (2026*c*) ii) Prior to 1993: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2022	Real net capital stock; 2017 SEK, millions The following series rescaled to match the end of year value of the series Eurostat (2026*h*) in 2017: i) Since 1993: Eurostat (2026*g*) ii) Prior to 1993: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1993: Eurostat (2026*b*) ii) Prior to 1993: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000: i) Since 1993: Eurostat (2026*f*) ii) Prior to 1993: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)

Variable	Range	Description
co2	1839–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Eurostat (2026*a*) in the total of Eurostat (2026*a*) and Eurostat (2026*e*) ii) Prior to 1995: Feenstra et al. (2015, LABSH)
tfpe	1950–2022	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2022	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

20. Switzerland (CH)

Variable	Range	Description
gdp	1950–2024	Real GDP; 2017 CHF, millions The following series rescaled to match the value of the series Bundesamt für Statistik (2026*e*) in 2017: i) Since 1996: Cumulative product of one plus the series Bundesamt für Statistik (2026*d*) divided by 100 ii) Prior to 1996: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RGDPNA)
nks	1950–2024	Real net capital stock; 2017 CHF, millions The following series rescaled to match the end of year value of the series Bundesamt für Statistik (2026*g*) in 2017: i) Since 1996: Bundesamt für Statistik (2026*f*) ii) Prior to 1996: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands The following series divided by 1000: i) Since 2010: Bundesamt für Statistik (2026*c*) ii) Prior to 2010: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions The following series divided by 1000000: i) Since 2010: Bundesamt für Statistik (2026*i*) ii) Prior to 2010: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1858–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*)

Variable	Range	Description
		ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1950–2024	Labor share i) Since 1995: Share of Bundesamt für Statistik (2026*a*) in the total of Bundesamt für Statistik (2026*a*), Bundesamt für Statistik (2026*h*) and Bundesamt für Statistik (2026*b*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, LABSH)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017

21. United Kingdom (GB)

Variable	Range	Description
gdp	1948–2024	Real GDP; 2017 GBP, millions The following series rescaled to match the value of the series Office for National Statistics (2026*c*) in 2017: Office for National Statistics (2026*b*)

Variable	Range	Description
nks	1950–2024	Real net capital stock; 2017 GBP, millions The following series multiplied by 1000 after being rescaled to match the end of year value of the series Office for National Statistics (2026*f*) in 2017: i) Since 1995: Office for National Statistics (2026*e*) ii) Prior to 1995: i) backdated using growth rates (previous period) of Feenstra et al. (2015, RNNA)
emp	1950–2024	Employment; persons, thousands i) Since 1980: OECD (2026*b*) ii) Prior to 1980: i) backdated using growth rates (previous period) of Feenstra et al. (2015, EMP)
hrs	1950–2024	Hours worked; millions i) Since 1980: OECD (2026*c*) ii) Prior to 1980: i) backdated using growth rates (previous period) of the product of Feenstra et al. (2015, EMP) and Feenstra et al. (2015, AVH)
co2	1751–2023	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1948–2024	Labor share Share of Office for National Statistics (2026*a*) in the total of Office for National Statistics (2026*a*) and Office

Variable	Range	Description
		for National Statistics (2026*d*)
tfpe	1950–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1950–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

22. United States (US)

Variable	Range	Description
gdp	1929–2024	Real GDP; 2017 USD, millions The following series rescaled to match the value of the series Bureau of Economic Analysis (2026*b*) in 2017: Bureau of Economic Analysis (2026*a*)
nks	1925–2024	Real net capital stock; 2017 USD, millions The following series rescaled to match the end of year value of the series Bureau of Economic Analysis (2026*d*) in 2017: Bureau of Economic Analysis (2026*c*)
emp	1948–2024	Employment; jobs, thousands Bureau of Labor Statistics (2026*a*) multiplied by 1000
hrs	1948–2024	Hours worked; millions Bureau of Labor Statistics (2026*b*) multiplied by 1000

Variable	Range	Description
co2	1800–2022	Carbon dioxide emissions; metric tons, thousands i) Since 1990: OECD (2026*a*) ii) Prior to 1990: i) backdated using growth rates (previous period) of CO2 in Hefner and Marland (2025)
sox	1850–2023	Sulfur oxides emissions; metric tons, thousands i) Since 1990: OECD (2026*d*) ii) Prior to 1990: i) backdated using growth rates (previous period) of SO2 in Smith et al. (2011)
ls	1948–2024	Labor share Bureau of Labor Statistics (2026*c*)
tfpe	1948–2024	Total factor productivity (employment-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
tfph	1948–2024	Total factor productivity (hour-based); Index (2017 = 1) Own calculations (see Appendix A) rescaled to match 1 in 2017
pop	1950–2025	Population, millions The following series divided by 1 000 000: i) Since 1960: World Bank (2026) ii) Prior to 1960: i) backdated using growth rates (previous period) of Feenstra et al. (2015, POP)

A. Construction of Employment- and Hour-Based TFP Indices

The employment- and hour-based TFP indices tfpe and tfph, respectively, are calculated based on gdp and a chained Törnqvist index of combined inputs in line with the TFP calculation of the Bureau of Labor Statistics (`https://www.bls.gov/opub/hom/opt/calculation.htm`) and the methodological note to experimental statistics on (crude) multifactor productivity on the corresponding eurostat webpage (`https://ec.europa.eu/eurostat/web/experimental-statistics/multifactor-productivity`). In short, given observations for $t = 1, 2, \ldots$, tfpe and tfph (prior to rescaling to match the value 1 in 2017) are calculated as follows:[1]

$$\begin{aligned}
\mathrm{tfpe}_t &\coloneqq \frac{\mathrm{gdp}_t}{\prod_{s=2}^{t} \exp\bigl((1 - w_{L,s})\Delta \ln(\mathrm{nks}_s) + w_{L,s}\Delta \ln(\mathrm{emp}_s)\bigr)},\\
\mathrm{tfph}_t &\coloneqq \frac{\mathrm{gdp}_t}{\prod_{s=2}^{t} \exp\bigl((1 - w_{L,s})\Delta \ln(\mathrm{nks}_s) + w_{L,s}\Delta \ln(\mathrm{hrs}_s)\bigr)},\\
w_{L,t} &\coloneqq \frac{1}{2}(\mathrm{ls}_{t-1} + \mathrm{ls}_t),
\end{aligned}$$

where the variables gdp, nks, emp, hrs, and ls are as described in the corresponding sections above.

[1]Note that empty products, i. e., here the denominators of tfpe_t and tfph_t for $t = 1$, evaluate to one.

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